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Palaeomagnetism of the Lower Carboniferous Billefjorden Group, Spitsbergen

D. R. WATTS

Department of Geology, The University, Glasgow, G12 8QQ, U.K.

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Abstract – Palaeomagnetic data from mid to late Palaeozoic rocks are potentially valuable for testing models of the assembly of Spitsbergen and for determining the palaeo-position of parts of Spitsbergen relative to the major tectonic elements that comprised the Old Red Continent. Fine-grained red sandstones from the Tournaisian to Namurian Billefjorden Group, collected by members of Cambridge Spitsbergen expeditions, were subjected to stepwise chemical and thermal demagnetization. The characteristic magnetization is found in normal and reverse polarities and corresponds to a pole position at 23° S, 332 °E, dp = 5.7°, dm = 10.5°, which is near an early Carboniferous pole computed for the Baltic Shield–Russian Platform (Baltica). When compared to Laurentia (North America) in the context of the revised Mauch Chunk palaeomagnetic study, Spitsbergen falls a few degrees south of the Bullard reconstruction but the error of the determination of the palaeolatitude overlaps with this position. Therefore relative motion between any part of Spitsbergen and Laurentia and Baltica during the Carboniferous is not resolved.

1. Introduction

The mid to late Palaeozoic palaeomagnetic record of the Atlantic-bordering continents may be used to determine the disposition of the tectonic elements related to the later stages of evolution of the Caledonian–Appalachian mountain belts. Of recent interest is the documentation of the extent and movement of displaced terrains within this belt (Kent & Opdyke, 1978, 1979; Van der Voo, French & French, 1979) and, on the larger scale, possible departure of the distribution of the major continental areas from the Bullard, Everett & Smith (1965) and similar reconstructions (Scotese *et al.* 1979; Van der Voo, 1983).

Lefort & Van der Voo (1981) and Van der Voo (1983) outlined a testable hypothesis, based on available palaeomagnetic data, of the disposition of three major tectonic elements that comprised the Old Red Continent. The elements are Laurentia (North America with the Labrador Sea closed), Baltica (Baltic Shield with the Russian Platform) and Armorica (southern Europe). According to this hypothesis, the three elements were joined during Old Red Sandstone time (late Silurian to late Devonian) but Baltica and Armorica were 15° to 20° south of the Bullard, Everett & Smith (1965) position. Baltica and Armorica then moved northward with respect to North America to the Bullard, Everett & Smith (1965) position during early Carboniferous time along strike slip faults. The driving force of this movement was the collision of Gondwanaland with the Old Red Continent. Slivers of continental crust with dimensions of the order of several hundred kilometres that were on the European side of the strike slip lineament(s) but remained attached to North America after the opening of the present Atlantic Ocean are now identified as displaced terrains such as Acadia (Kent & Opdyke, 1978).

As argued by Harland (1978), Harland & Wright (1979), and Harland (1980), the idea of sinistral strike slip motion displacing slivers of continental crust during the late stages of the Caledonian orogeny is an attractive hypothesis that explains many aspects of the geology of Spitsbergen. Spitsbergen may be divided into at least three provinces of older rocks which are separated by two major fault zones. The eastern province has strong similarities to central eastern Greenland (Harland, 1978; Harland & Wright, 1979). The eastern major fault zone is identified as the Billefjorden Fault zone (Fig. 1) which separates metamorphosed Precambrian rocks from unmetamorphosed Old Red Sandstone. The Old Red Sandstone and the metamorphic rocks have both undergone late Devonian deformation, the Svalbardian orogeny, which is of a style consistent with a component of strike slip motion and a component of compression (transpression). Harland & Wright (1979) suggest that the three provinces of Spitsbergen may have moved as much as 2000 kilometres relative to each other, with the eastern province initially the furthest south.

Lower Carboniferous fluvial sediments unconformably overlie the rocks folded during the Svalbardian event and are found on both sides of the Billefjorden Fault zone (Fig. 1). If the strike slip component of movement was associated with the Svalbardian folding, then the movement must have

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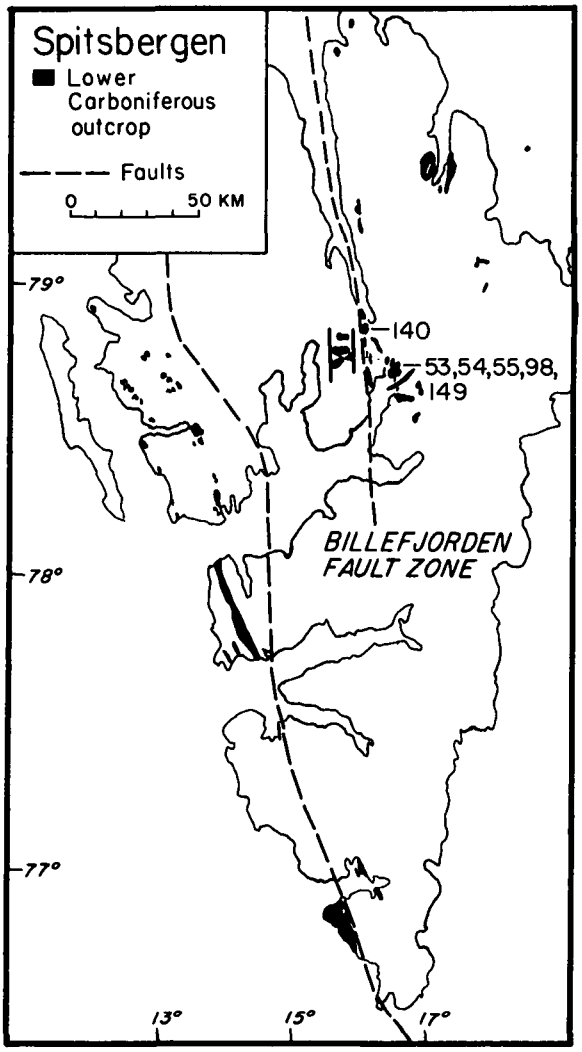


Figure 1. Geological map of Spitsbergen, after Cutbill & Challinor (1965), showing areas of outcrop of Lower Carboniferous sediments, major faults, and the palaeomagnetic sampling sites.

been pre-early Carboniferous time, in contrast with the hypothesis of Lefort & Van der Voo (1981) and Van der Voo (1983) which suggests that large-scale motion took place during early Carboniferous time.

Roy & Morris (1983) disputed the idea that substantial Carboniferous relative movement between Laurentia and Acadia occurred. They observe that the relevant palaeomagnetic data taken to be representative of Laurentia for early Carboniferous strata are dominantly of reversed polarity. Magnetization found in the early Carboniferous rocks of the Canadian basin in Acadia is of normal and mixed polarity. This implies that the palaeomagnetic data used to determine an early Carboniferous position for Laurentia may not be representative of the same age as the data from Acadia, and a comparison may be invalid. Indeed, the predominance of reversed polarity in the data from Laurentia suggests that the magnetization was acquired during the Kiaman reversed epoch and is of late Carboniferous to Permian age. Therefore Roy & Morris (1983) believe that the apparent Carboniferous displacement of Acadia with respect to Laurentia is only the consequence of widespread Kiaman age remagnetization of the North American craton. By implication, early Carboniferous results from Acadia are, in fact, representative of Laurentia.

Kent & Opdyke (1984) support this view as a result of further work on the Mauch Chunk Formation of the Appalachians. They show that the pre-folding magnetization observed by Knowles & Opdyke (1968) is the result of a pre-folding characteristic magnetization and a syn-folding overprint. The characteristic magnetization is of mixed polarity, contrasting with other results (mainly from limestones) used to position Laurentia which are of reversed polarity. The characteristic magnetization yields a pre-folding direction with an associated palaeolatitude which is consistent with the palaeolatitudes deduced from the early Carboniferous rocks of the Acadia terrain and requires no relative movement.

When compared to the European record, the Mauch Chunk revised result is consistent with a Bullard, Everett & Smith (1965) style reconstruction rather than a southerly position of Baltica and Armorica (Kent & Opdyke, 1984). This implies that if large-scale strike slip motion occurred, it must be older than early Carboniferous.

Detailed palaeomagnetic data from Spitsbergen would allow a test and further clarification of these ideas. Displacement of terrains within Spitsbergen may well be independent of displacement elsewhere in the Old Red Continent. As the northern extremity of the Old Red Continent and the Caledonian orogen, the detection of movement (or the lack of it) with respect to the major elements of the Old Red Continent will provide constraints on hypothesis of the late stages of evolution of the Caledonide–Appalachian mountain belt.

Oriented samples for palaeomagnetic study were collected by the Cambridge Spitsbergen expedition group during the late 1950s (Harland, 1959). The natural remnant magnetization (NRM) and the results of isolated, single step alternating field demagnetization experiments were determined by Bidgood (private communication). Included in this study were samples from the Lower Carboniferous Billefjorden Group collected from localities east of the Billefjorden Fault Zone (Fig. 1). Reported here are the results of chemical and thermal demagnetization of these samples, the mean palaeomagnetic direction and its implication for the problems outlined above.

2. Geological setting of the Lower Carboniferous palaeomagnetic samples

Figure 1 is a geological map (after Cutbill & Challinor, 1965) showing the outcrop location of Lower Carboniferous sediments in Spitsbergen, the Billefjorden Fault Zone and the palaeomagnetic collecting

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localities. All samples are from east of the Billefjorden Fault Zone and from the upper portions of the Billefjorden Group (Bidgood, private communication).

The Billefjorden Group consists predominantly of fluvial sediments, up to 200 metres thick, that include fine-grained red sandstones. On the basis of palynology, the group may range in age from Tournaisian to Namurian times. The sediments are flat-lying, gently tilted at some localities by normal faulting, and overlie with a major angular unconformity the rocks folded during the Svalbardian event. The Billefjorden Group was formerly known as the Billefjorden Sandstone Formation and the Culm (Bidgood, private communication).

The oriented samples investigated in this study were collected from the upper portions of the Billefjorden Group. The samples are fine-grained red sandstones. All samples were oriented with a rock compass (Bidgood & Harland, 1959).

3. Palaeomagnetic properties and directions

Magnetic moments were measured at the University of Leeds using a fluxgate spinner magnetometer and the Leeds cryogenic magnetometer. Chemical demagnetization was carried out by cutting slots in the specimens perpendicular to bedding and immersing the samples in 8N HCl for periods of from 4 to 60 h. After each immersion, the samples were washed in distilled water, dried and then measured. The entire operation was carried out in the Earth's magnetic field. Thermal demagnetization was done using a bench-type device with the ambient field annulled to 5 nT by nested mu-metal shields. Alternating field demagnetization up to 200 nT was found to remove only low-coercivity components and the characteristic magnetization was not fully revealed. Hence, only thermal and chemical demagnetizations were carried out on the available specimens. After each chemical or thermal demagnetization step, the results were plotted on orthogonal projections (Zijderveld, 1967) to evaluate the effect of the treatment and to determine the next step.

Figure 2 shows orthogonal projections of typical demagnetization results. A soft component, with steep positive inclination, was removed during the early stages of treatment to reveal a characteristic magnetization with a thermally distributed blocking temperature spectrum. The characteristic magnetization was typically removed by 680 °C or by leaching up to 150 h in 8N HCl. The chemical demagnetization results are rather noisy (Fig. 2a) compared to the thermal demagnetization results (Fig. 2b, 2c, 2d), possibly because the leaching and drying were carried out in the Earth's magnetic field.

The direction of the characteristic magnetization was estimated by vector subtraction of the two endpoints that best fit the decay trajectory. Samples

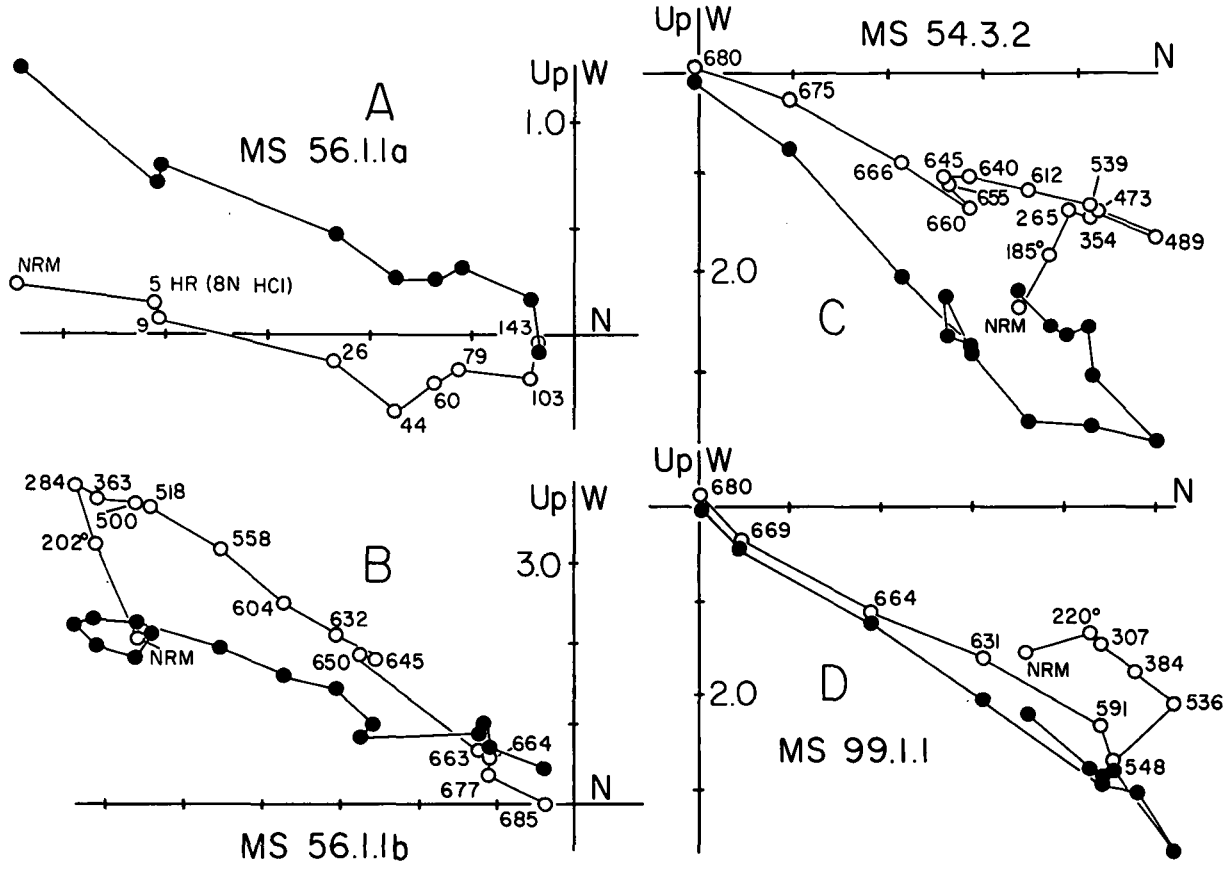


Figure 2. Orthogonal projections (Zijderveld, 1967) of typical demagnetization results. The open circles denote projection on the vertical plane. Closed circles denote projection on the horizontal plane. The axes are marked in units of 10^{-3} amp/metre.

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Table 1. Palaeomagnetic directions, statistics and pole positions

Site	N	Dec.	Inc.	k	Alpha 95	Pole Lat.	Pole Long.	dp	dm
54	8	52°	23°	11.2	17.3°				
55	1	39°	52°	—	—				
56	6	211°	−24°	8.3	24.7°				
98	3	34°	27°	54.4	16.9°				
140	2	226°	−26°	22.0	—				
149	1	218°	−48°	—	—				
Site-mean	6	40°	33°	30.8	12.3°	−27°	333°	8.0°	14.0°
Sample-mean normal	12	46°	27°	12.8	12.6°				
Sample-mean reversed	9	210°	−30°	5.2	24.9°				
Sample-mean	21	42°	27°	11.8°	9.6°	−23°	332°	5.7°	10.5°

Site numbers are as given by Bidgood (private communication); N is the number of samples used to compute the mean direction; Dec. is the mean declination; Inc. is the mean inclination; k is the estimate of Fisher's (1953) precision parameter; Alpha 95 is the apical half angle of 95% confidence (Fisher, 1953); pole lat. is the latitude of the pole position; Pole long. is the longitude of the pole position; dp and dm are the semi-axis of the oval of 95% confidence (Creer, Irving & Runcorn, 1957).

exhibiting erratic, random demagnetization trajectories were rejected as uninterpretable.

The site-mean directions, two-tier (mean of the site means) and the sample-mean directions of the characteristic magnetization are given in Table 1 with the associated pole positions and statistics. Because of the different numbers of samples available from each site, the sample-mean direction is perhaps the more valid, as the two-tier mean gives equal weight to sites represented by only one sample.

Both normal and reversed directions of magnetization are recorded in these rocks. The mean of the normal directions is antipodal with the mean of the reversed directions (Table 1). Individual sites were either completely normal or completely reversed. There is no apparent difference in the blocking temperature spectra or the resistance to chemical leaching between the normal or reversed magnetization. The pole derived from the sample-mean characteristic magnetization direction is lat. = 23° S, long. = 332° E, dp = 5.7°, dm = 10.4°.

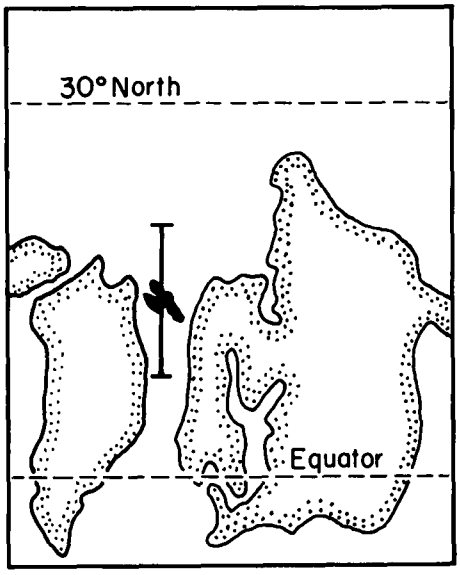


Figure 3. Early Carboniferous palaeogeographic reconstruction of Laurentia, Baltica and Spitsbergen consistent with the available palaeomagnetic data. The error of the estimate of the palaeolatitude of Spitsbergen is indicated by the bar.

4. Comparison with other early Carboniferous palaeomagnetic results

Scotese *et al.* (1979) calculated a mean early Carboniferous pole for Baltica from data derived primarily from the Russian Platform. This pole falls, in the European frame of reference, at 28° S, 344° E, near the pole derived from the Billefjorden Group.

Bachtadse, Heller & Kröner (1983) derived a palaeomagnetic pole from Lower Carboniferous sedimentary rocks of Hercynian Europe at 41° S, 342° E. The disagreement with the Scotese *et al.* (1979) pole cannot be entirely explained by local tectonic rotation as the structurally corrected inclinations from the Hercynian results imply a more northerly palaeolatitude (13°). There are a number of plausible explanations for the difference. The Bachtadse, Heller & Kröner (1983) result is based on only two sites and is statistically limited. The data from the Russian Platform are not documented with details of the physical properties of the magnetization or the experimental technique. The possibility of remagnetization such as affects the North American record cannot be discounted. The Scotese *et al.* (1979) pole is used to derive a position for Baltica in Figure 3 only because it is based on a substantially larger data set.

The problems of the North American Carboniferous record were reviewed above. The position of North America in Figure 3 is that produced by Kent & Opdyke (1984) based on the revised Mauch Chunk result. The position of Spitsbergen in Figure 3 is derived from the assumptions that it is near the Greenland margin during the early Carboniferous and that the sample-mean palaeomagnetic pole in Table 1

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is more valid than the pole calculated from the site-mean direction of magnetization. The error of the estimate of the palaeolatitude, dp, is shown by the solid bar.

Although the palaeomagnetic results give a palaeolatitude rather south of the Bullard, Everett & Smith (1965) position, the error bar overlaps with such a position. Also, if the site-mean average direction is used to derive a position for Spitsbergen, it would be closer to the Bullard, Everett & Smith (1965) position. Strictly speaking, this result is valid only for the part of Spitsbergen east of the Billefjorden Fault zone.

5. Summary and conclusions

Thermal and chemical demagnetization reveal a characteristic magnetization of the fine-grained red sandstones of the Lower Carboniferous Billefjorden Group of Spitsbergen. Normal and reversed polarity occurs in approximately equal frequency. The sample-mean direction is consistent with a palaeolatitude of 14° N ±5.7°. The site-mean direction yields a somewhat higher palaeolatitude at 18° N.

When compared to Baltica and Laurentia in Figure 3 using available palaeomagnetic data, Spitsbergen appears somewhat south of the Bullard, Everett & Smith (1965) position, but this position is within the limit of the estimate of the error in palaeolatitude. A position consistent with 2000 km of sinistral strike slip motion with respect to Laurentia is outside this error bar when the position for Laurentia is derived using the revised Mauch Chunk Formation result (Kent & Opdyke, 1984) and the data from the Canadian Carboniferous basin.

The mixed polarity of the Billefjorden result implies an age of magnetization older than the Kiaman reversed interval and argues against Permo-Carboniferous remagnetization. The case for comparison with the revised Mauch Chunk result and the Canadian basin poles is therefore strengthened.

The error of the palaeolatitude estimate does not preclude strike slip or other relative motion of the order of several hundreds of kilometers. As the error overlaps with the Bullard, Everett & Smith (1965) position, it is appropriate to say that post early Carboniferous relative movement is not resolved. Further work to reduce the error of the palaeolatitude determination and to extend the geographic sampling base is required.

Although it appears that large-scale strike slip movement later than early Carboniferous is inconsistent with the latest palaeomagnetic data, a case can still be made for such movement during the Devonian (Kent & Opdyke, 1984). This situation is compatible with Harland's (1980) hypothesis that large-scale movement between parts of Spitsbergen occurred during the Devonian and was complete by early Carboniferous time. However, as Kent & Opdyke (1984) point out, the remagnetization problem cannot yet be resolved for the Devonian as the positive fold test from the Catskill Formation (Van der Voo, French & French, 1979) only demonstrates that the magnetization is pre-Permian.

The tectonic setting of the Old Red Sandstone in Spitsbergen will allow a unique test of the age of magnetization of this unit as it was folded during the Devonian Svalbardian orogeny. The fold test applied to this unit will unequivocally distinguish between Devonian and later magnetizations and contribute to the resolution of this problem.

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